FACTORIZATION THROUGH THE RESIDUAL SPACE FOR DECENT SPACES WITHOUT THE NOETHERIAN HYPOTHESIS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Noetherian factorization. lemma-factor-through-residual-space-Noetherian (carded L26727). The model case.
- Decency local on the target. lemma-etale-named-properties (carded L26706). Decency pulls back along etale morphisms.

Source reference. Original lemma 0BBN: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1750–1750. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0H1R: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3309–3309. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H1M: focused-seeds/geometry/S9-decent-factor-through-residual-space-decent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IE: focused-seeds/geometry/S9-decent-factor-through-residual-space-decent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JJ: focused-seeds/geometry/S9-decent-factor-through-residual-space-decent/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $x \in |X|$ be a point. Assume*

(1) *$|f|(|Y|)$ is contained in $\{x\} \subset |X|$,*

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(2) *Y is reduced, and*
(3) *x can be represented by a quasi-compact monomorphism $x : \mathrm{Spec}(k) \to X$ where k is a field (for example if X is decent).*

Then f factors through the residual space $Z_x = \mathrm{Spec}(k)$ of X at x.

Proof. By Lemma 0H1R we have $Z_x = \mathrm{Spec}(k)$.

Preliminary remark: since $\mathrm{Spec}(k) \to X$ is a monomorphism, it suffices to find a surjective étale morphism $Y' \to Y$ such that $Y' \to X$ factors through Z_x. A remark here is that Y' is reduced as well.

After replacing X by a quasi-compact open neighbourhood of x, we may assume X quasi-compact. By Lemma 0BBN, x is a point of $T \subset U \subset X$ where $T \to U$ (resp. $U \to X$) is a closed (resp. open) immersion, and T is a scheme. By Properties of Spaces, Lemma 03IE, f factors through U, so we may assume $U = X$. Then f factors through T because Y is reduced, see Properties of Spaces, Lemma 03JJ. So we may assume that $X = T$ is a scheme. By our preliminary remark we may assume Y is a scheme too. This reduces us to Morphisms, Lemma 0H1M. □